BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: *The Smart Mobile Plant-Health Sentinel*

by

Group 8

Section 4

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Abstract

This project is about plant health and minimising water wastage and human work. In this project we present the design and implementation of a Smart Mobile "Plant-Health" Sentinel, which is an autonomous robotic system developed to monitor and maintain plant health. The robot navigates through a path using plant position markers and it does soil moisture analysis at each plant. Based on plant moisture reports, the system makes intelligent decisions. Additional features such as watering duration, water tank level monitoring and obstacle detection that ensures safe and efficient operation.This project reduces water wastage and with it human work is going to get minimized.

Keywords

Autonomous robot, Soil moisture sensors, Smart Decision, Arduino, Precision agriculture

1. Introduction

In urban areas many people desire to have a garden of their own but they don't have the time to take care of that garden and maintain the plant's health . For that we are bringing Smart Mobile Plant-Health Sentinel. This project is all about Smart Mobile Plant-Health Sentinel's design and implementation . The objective of this project is to develop an autonomous robot which can move through a garden, check soil conditions, and water plants when it is needed.This project talks about computer interfacing as it uses sensors, actuators, and a microcontroller to create a real - time automated system. It combines sensing, decision-making, and actuation into one complete robot.

2. Related Work/ Inspiration

Several automated robotic systems exist, but most of them rely on fixed sensors and provide generalized watering. They don't check the soil and water plants when it is needed.

Our project is different in many ways such as:

- It isn't static, it is totally autonomous
- It checks each and every plant's moisture thresholds and waters accordingly.
- It introduces watering duration and allows rechecking.

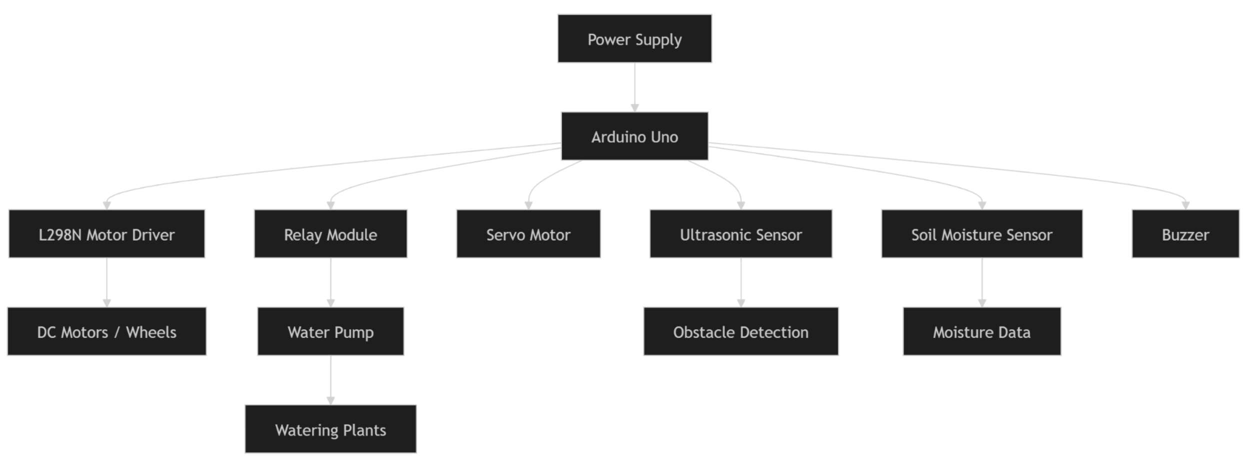
This makes our system more accurate and practical for real-world use.

3. Technical Approach

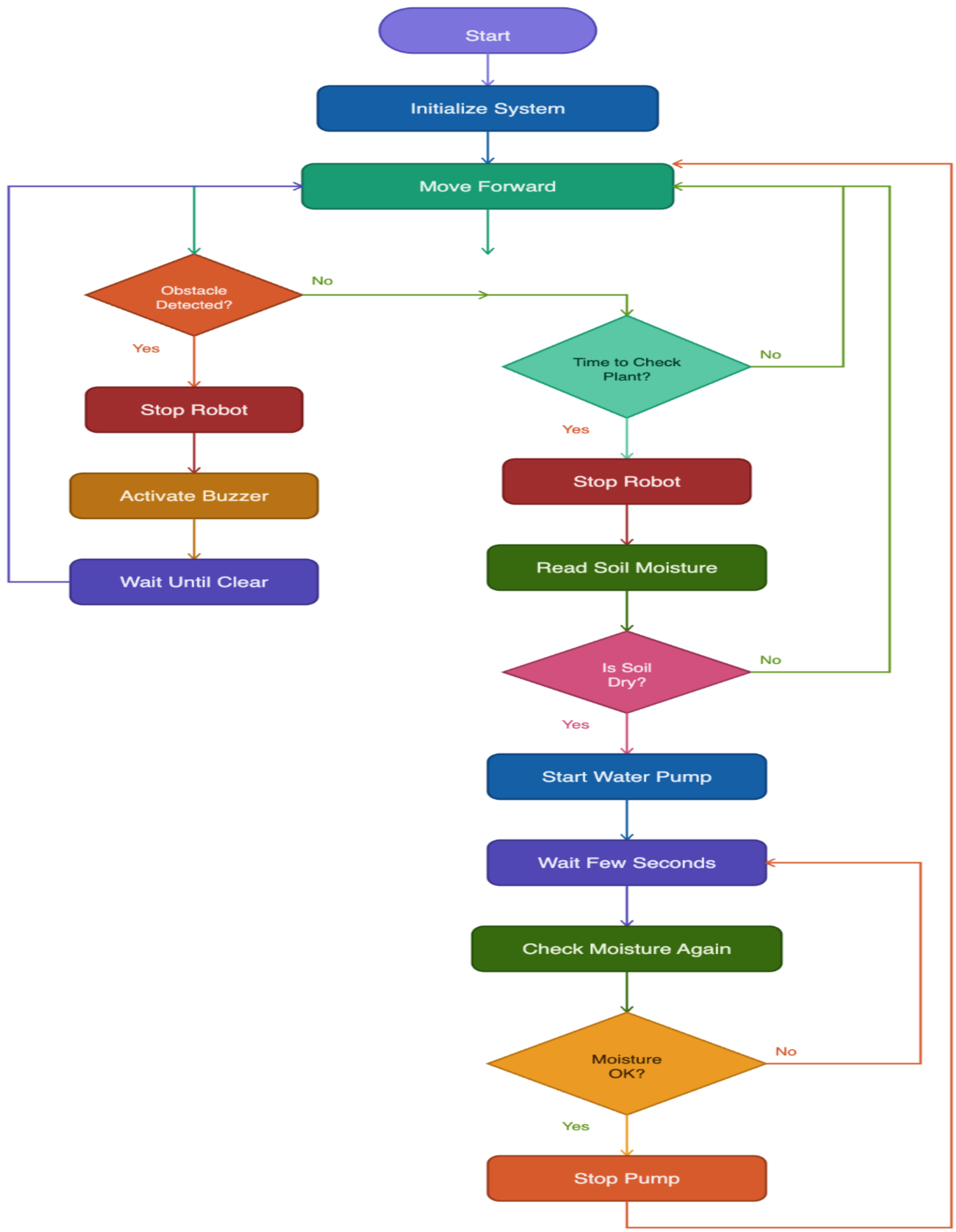
3.1 System Architecture:

In our Project we used an ultrasonic sensor to search for an obstacle and if there is any obstacle it has a buzzer which will start buzzing. The soil moisture sensor is used to ensure accurate watering. When the robot reaches a plant, it initially gives a small amount of water and then measures the soil moisture level with the sensor. If the sensor detects that the soil has reached the required moisture threshold, the system stops watering immediately. However, if the soil is still below the level after the watering, the robot starts to supply water in small amounts until the appropriate moisture condition is gained. To tackle overwatering we have set maximum time watering.

Block Diagram



Flowchart


Start
Initialize System
Move Forward
Obstacle Detected?
No
Yes
Stop Robot
Activate Buzzer
Wait Until Clear
Time to Check Plant?
No
Yes
Stop Robot
Read Soil Moisture
Is Soil Dry?
No
Yes
Start Water Pump
Wait Few Seconds
Check Moisture Again
Moisture OK?
No
Yes
Stop Pump

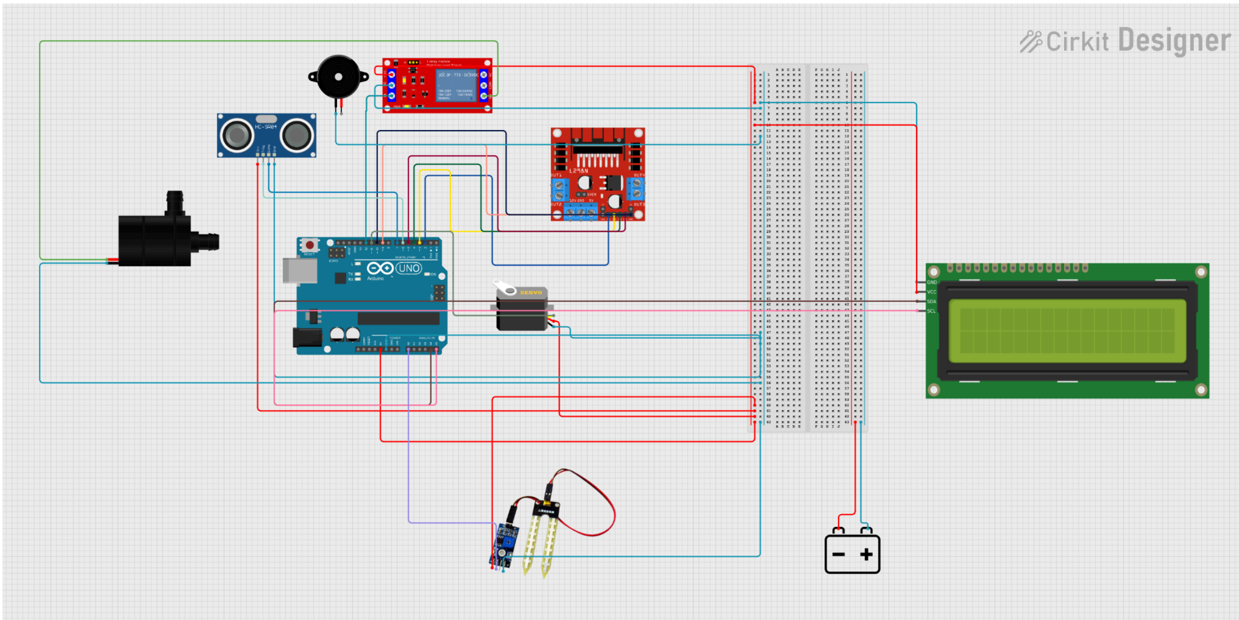
3.2 Components Used:

- Arduino Uno R3
- Soil Moisture Sensor
- Ultrasonic Sensor
- Servo Motor (SG90, 180° Rotation)
- Submersible 3V Mini DC Water Pump
- Relay Module (1 Channel, 5V)
- L298N Motor driver
- Water Pipe (8×6 mm, 1 ft)
- 18650 Battery Holder (3-cell)
- 18650 Lithium-ion Batteries (3000mWh)
- DC Motor (2x)
- Car chassis with castor wheel

3.3 Cost Breakdown:

Component	Quantity	Unit Price (৳)	Total (৳)
Arduino Uno R3	1	750	750
Soil Moisture Sensor	1	90	90
Ultrasonic Sensor (HC-SR04)	1	145	145
Servo Motor (SG90)	1	135	135
L298N Motor Driver	1	219	219
Submersible Mini DC Water Pump (3V–6V)	1	120	120
Relay Module (1 Channel, 5V)	1	95	95
Water Pipe (8×6 mm, 1 ft)	1	20	20
18650 Battery Holder (3-cell)	1	75	75
18650 Battery Holder (2-cell)	1	48	48
18650 Li-ion Battery (3000mAh)	5	120	600
2 Wheel Robot Car Chassis	1	550	550
Grand Total			**৳ 2,847**

3.4 Circuit/Schematic:



Functionality:

Autonomous movement and Navigation:

The robot moves and navigates safely while it detects obstacles in its path. After 10 seconds, the robot will stop and check the moisture sensor functionality. After 30 seconds, the robot will turn around.

Obstacle detection and warning

The ultrasonic sensor is used to scan the entire environment to detect obstacles. The control logic simultaneously calculates the distance to objects. If anything is detected, the actuator logic properly stops the robot from moving and other functionalities. The buzzer will make a sound while the robot is stopped. Once the road is clear, it starts again, and its workflows resume.

Feedback-driven Precise Irrigation

The watering is done with a process where the water is given based on the soil/plant needs.
When the robot comes to a plant near the soil, it does a sensing and actuation task. The soil moisture sensor checks the moisture level. There is a threshold. If the soil moisture sensor senses and compares with the threshold. If it is more than the threshold, it means the soil is dry and requires water. Then the robot will provide water through the 3V DC water pump. After a few

seconds, it will again sense, measure the moisture level and see if the plant needs more water. If yes, then again give water, move forward. The moisture level is seen on the I2C.

Challenges:

The first problem we faced was the design and the placement of multiple components and sensors. We faced a lot of challenges, including the readings varying from the soil and other sensors . Obstacle detection is also very difficult, as small objects don't always get detected properly, it is affecting the navigation. Additionally, real-time processing of sensor data requires great programming to avoid delays. Achieving water control is also challenging as overwatering or under-watering will occur if the calculation is not perfect. Also the robot's movement was a challenging issue for us as the breadboard wasn't fitting on the car. Finally, making all hardware and software components work smoothly needs careful design and repeated testing to ensure the system works well.

4. Sustainability & Impact

4.1 Sustainability

- It reduces water wastage through putting in a small amount of water.
- It uses low-power components
- Saves time and minimises human work

4.2 Impact

- It is very useful for urban gardening, as urban people have less time to invest in it.
- It reduces human labour
- It helps beginners maintain plants easily and know about the soil.

4.3 Future Work

- May invent a mobile app to monitor the plants.
- Will use AI for plant disease detection.
- Will add GPS-based navigation for larger farms and gardens.

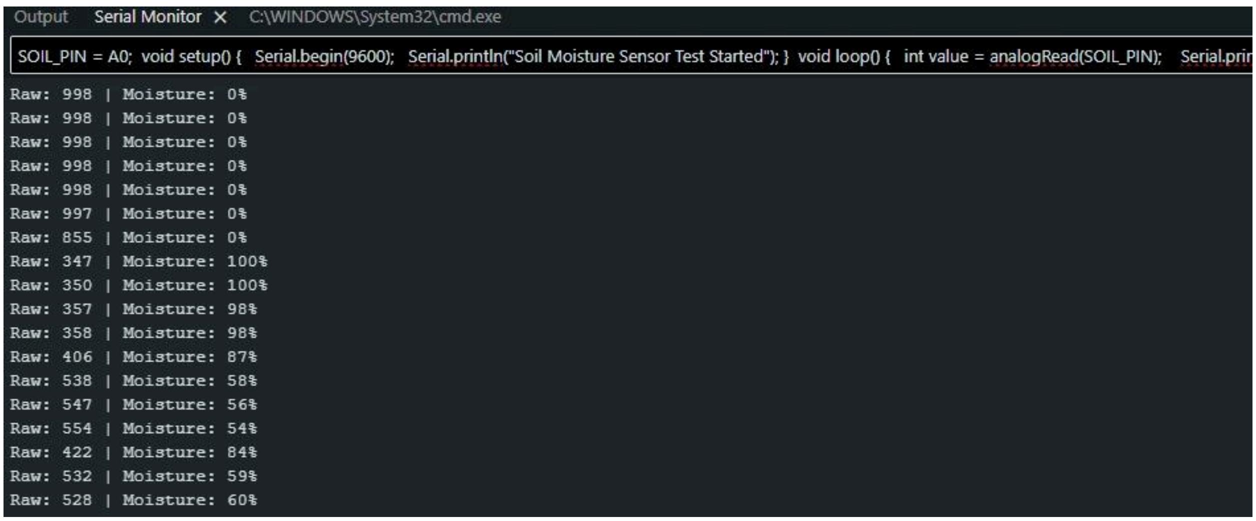
4.4 Limitations

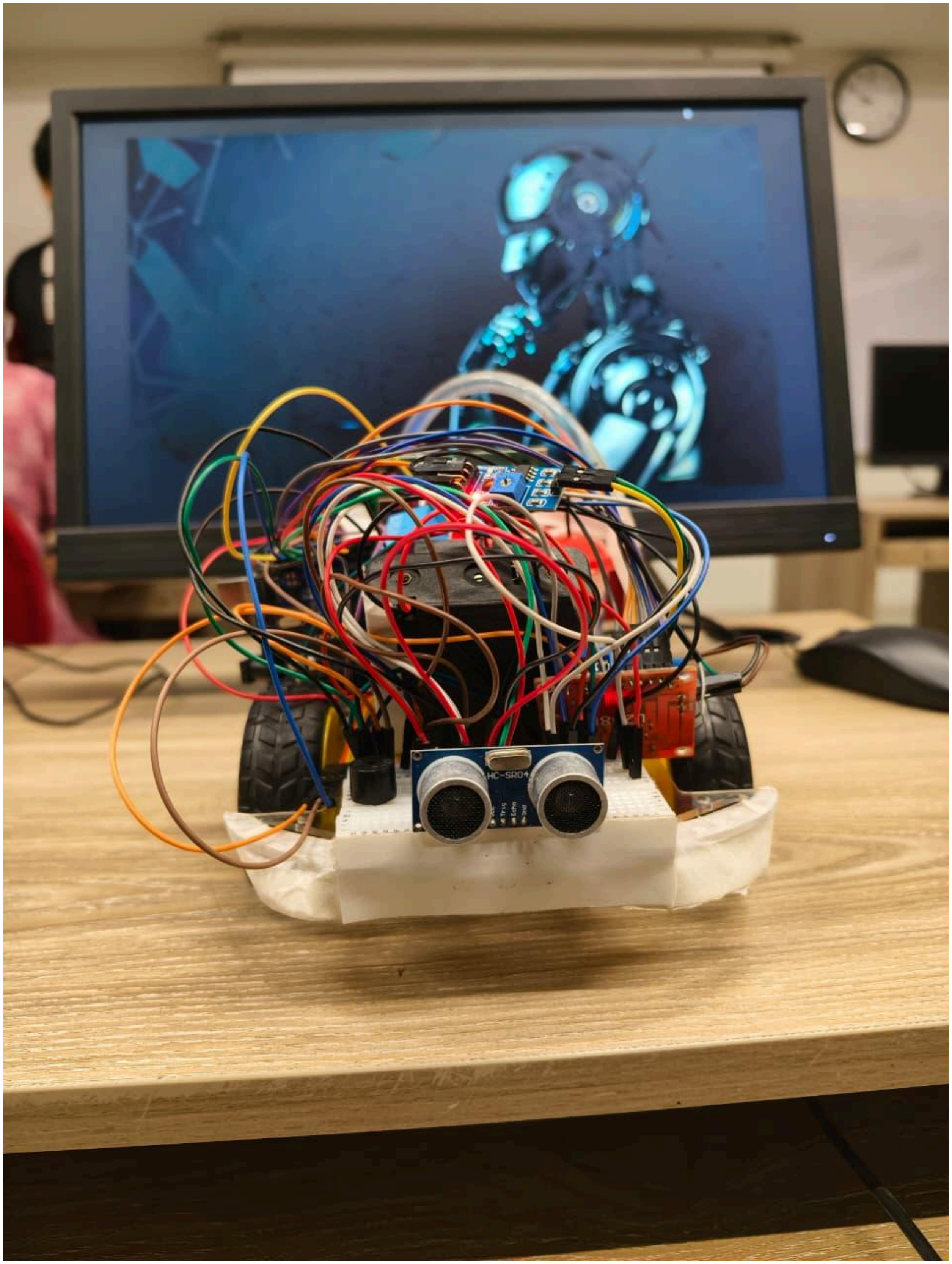
- We had limited small areas to work with.
- Sensor accuracy may vary, and it makes our work very difficult.

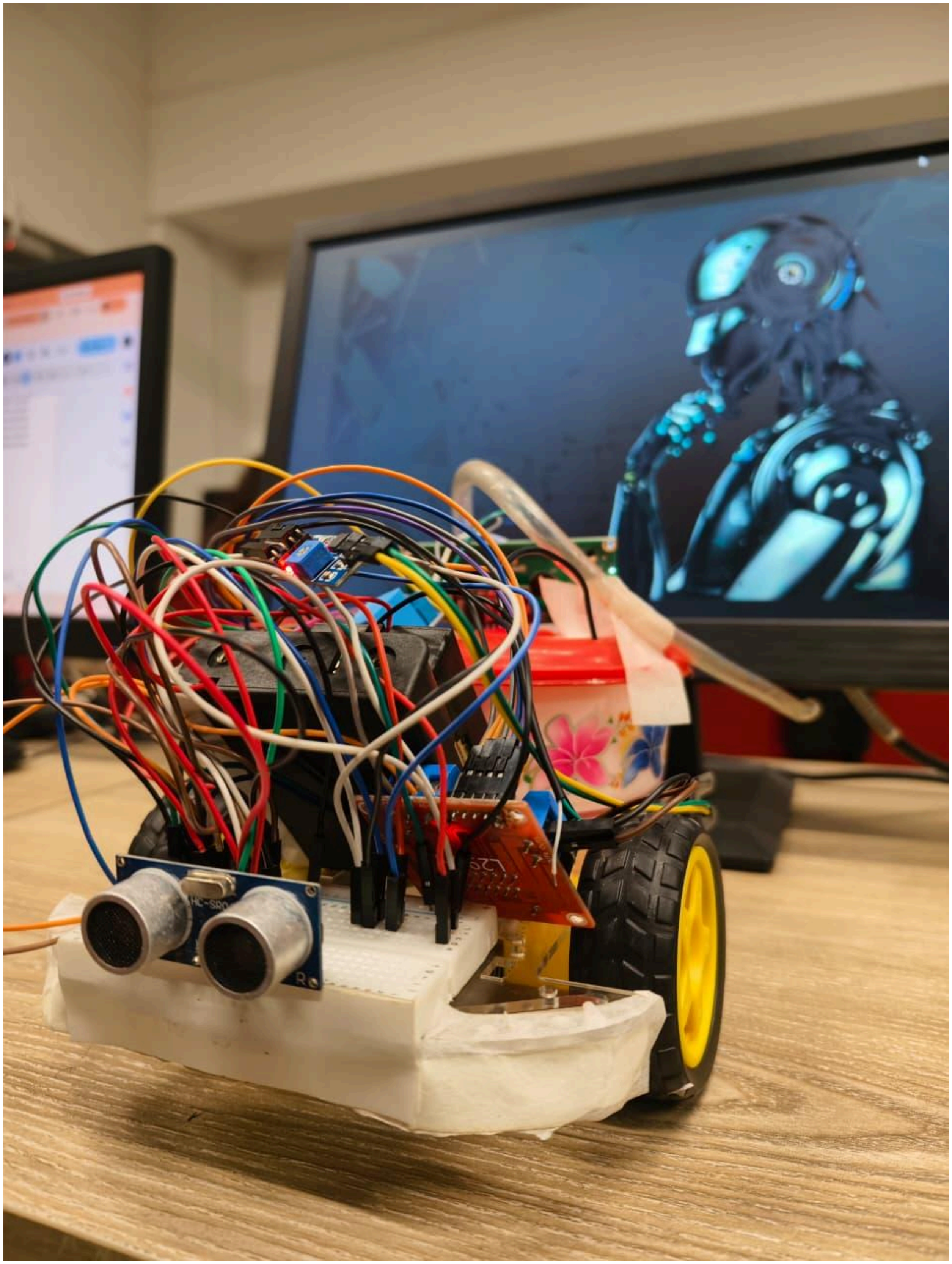
5. Results & Discussion

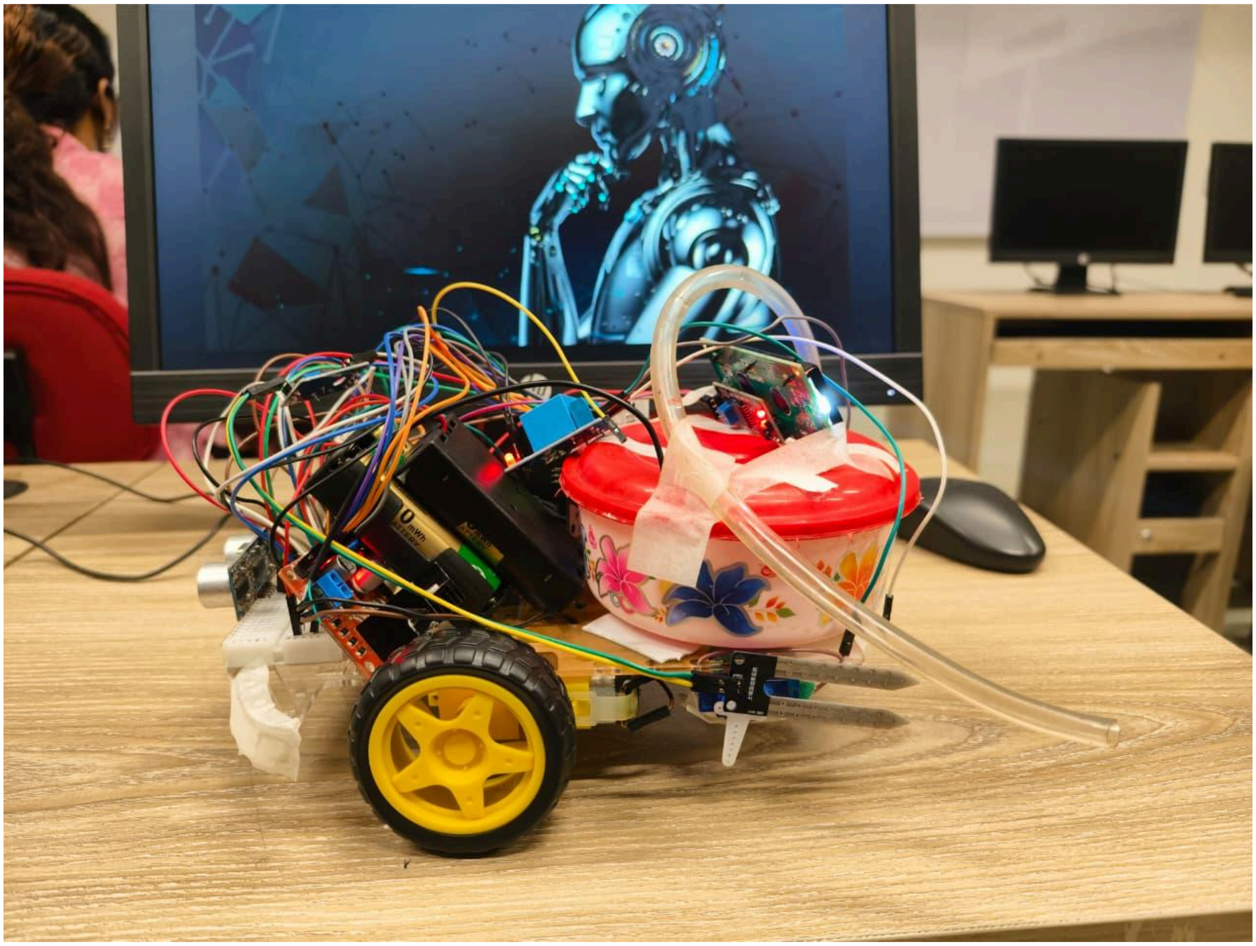
The robot works very well and completes all its main tasks properly. It can move from one plant to another, while moving, it can detect obstacles and avoid them, and that makes this robot safe to work. For every plant, the robot provides a small amount of water. After that, it checks the soil with a soil moisture sensor and compares it with a threshold and decides whether the plant needs further water or not. Based on that, it gives water in a controlled way, only as much as is needed.

In our testing time, we saw that our robot uses water in such a way that it uses less amount of water to avoid wasting water.




HC-SR04





6. Conclusion

The Smart Mobile Plant-Health Sentinel elaborates on how robotics can add an efficient touch to plant care through automation. With merging mobility, sensing, and intelligent decision-making, the robot provides results that are more precise and helps to take care of the plants more. Furthermore, the robot ensures the smart use of water and prevents water wastage.

This project highlights the importance of sustainability and shows how simple robotic systems can solve real-world problems.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Shiva Prasad Sarkar	23101302	Hardware Lead	Designed circuit and asselmbled, connected L298N motor driver with arduino and battery pack. Also connected relay module and water pump.
Rudaba Tabassum	22299125	Documentation & Testing	Wrote the project report, prepared and managed components, connected servo motor with soil moisture sensor and compiled the result.
Debashish Chowdhury	22201220	Sensor & Integration	Calibrated ultrasonic sensor for make during autonomous vehicle movement, and designing buzzer logic.
Md. Affan Hossain Rakib	22201936	Software Lead	Wrote the Arduino program for soil moisture threshold and water control from the submersible water pump.

References

Arduino. (2024). *Arduino Uno Rev3 documentation*. Arduino Docs. https://docs.arduino.cc/hardware/uno-rev3/

SparkFun Electronics. (n.d.). *Soil moisture sensor hookup guide*. https://learn.sparkfun.com/tutorials/soil-moisture-sensor-hookup-guide

Sundari, S., & Shanthi, P. (2021). *IoT based smart agriculture system using soil moisture sensor and Arduino*. Journal of Physics: Conference Series, 1911(1), 012014. https://doi.org/10.1088/1742-6596/1911/1/012014